

Assignment - 1

Title:

Handwritten Signature Verification Using Improved Combined Features: A Comprehensive Review

1. Introduction

Handwritten signatures are one of the most traditional and widely accepted forms of personal identification. Whether it is signing a cheque, authorizing a legal document, or verifying identity in administrative records, signatures continue to play a critical role in confirming a person's authenticity. However, with the rise in sophisticated forgery techniques, relying solely on visual inspection or basic image comparison for signature verification is no longer sufficient. Forged signatures can lead to significant financial fraud and legal issues, particularly in sensitive domains such as banking, insurance, and governmental services.

To combat this, the field of biometric authentication has increasingly focused on automating the process of signature verification. Signature verification methods can be broadly categorized into two types: **offline verification**, which uses scanned or imaged signatures, and **online verification**, which captures real-time data such as pen pressure, speed, and trajectory while signing. Both approaches have their advantages, but each alone has limitations when it comes to accuracy, especially in scenarios where only a small number of genuine samples are available for training.

The research paper titled "*Handwritten Signature Verification Method Based on Improved Combined Features*" addresses this limitation by proposing an innovative solution. It introduces a **hybrid verification system** that leverages the capabilities of both offline and online signature data. This dual approach not only enhances the richness of the captured signature data but also improves verification performance. By using a smart pen equipped with a pressure sensor and camera, both the image and dynamic motion of a signature are captured in real-time. The paper then uses **Support Vector Machine (SVM)** for processing the image (offline data) and **Dynamic Time Warping (DTW)** for analyzing motion data (online data). These two classifiers work independently, and their decision scores are intelligently combined using a proposed **Score Fusion based on Accuracy (SF-A)** method.

This integrated framework provides a more comprehensive and secure approach to handwritten signature verification, aiming to reduce both false acceptances (forgeries passing as genuine) and false rejections (genuine signatures being marked as fake).

2. Literature Review

Over the years, various researchers have attempted to improve the accuracy and reliability of signature verification systems. One of the key areas of focus has been **feature extraction**, where meaningful characteristics from a signature—such as its texture, shape, orientation, and timing—are converted into numerical representations for analysis.

Some researchers like Faiza et al. developed a method that combines **Gray-Level Co-occurrence Matrix (GLCM)** features with geometric data and then uses a feature selection algorithm called SKCPCA to refine the input to the classifier. Their approach reduced the False Acceptance Rate (FAR) significantly but risked losing useful data during the feature reduction process. Others like Bhunia et al. combined **wavelet features** with **Local Quantized Patterns (LQP)** to create multiple models for each user, using one-class SVMs to classify signatures. This method showed good results across multiple datasets but required significant computational resources.

More recently, deep learning methods have been applied, such as using **Convolutional Neural Networks (CNNs)** to automatically learn discriminative features from signature images. For example, Hafemann et al. achieved high accuracy by training CNNs on large datasets, while others proposed new network structures like LS2Net to handle varying signature sizes and improve performance on small samples. However, these methods often require large training datasets and powerful hardware, making them less practical in real-world settings where only a few genuine signature samples are available.

Regarding **classification**, traditional methods like **SVM**, **Random Forest**, and **Hidden Markov Models (HMM)** remain popular due to their balance of interpretability and performance. Distance-based approaches such as **DTW** have also been effective, particularly for online signatures, as they can measure similarity based on time-series alignment. However, each classifier has its strengths and weaknesses depending on the data type.

The idea of **combining features and classifiers** is not new but still evolving. Previous fusion methods have included score-level fusion, feature-level fusion, and decision-level fusion. Yet, most of them treat offline and online data separately. The reviewed paper distinguishes itself by proposing a new accuracy-weighted score fusion method (SF-A), which considers both types of data and intelligently merges the results from different classifiers based on their reliability.

3. Methodology

The core contribution of the paper is a multi-stage framework that combines offline and online signature data using improved feature extraction and classification techniques. The methodology consists of the following components:

a) Data Acquisition

The study uses a smart pen with a built-in camera and pressure sensor to simultaneously collect offline images and online dynamic data of handwritten signatures. A dataset was created consisting of **1200 signature samples** from **20 individuals**, with each person providing **30 genuine signatures** and **30 forged signatures**. The forgeries were obtained from 2–3 individuals who practiced imitating the signatures after being shown the originals, making the dataset more realistic and challenging.

b) Pre-processing

Offline images were processed through several steps to improve quality and uniformity:

- **Grayscale conversion**
- **Binarization**
- **Normalization** (resizing to 64x64 pixels)

These steps help reduce noise and make feature extraction more effective. Online data, such as pen pressure, x-y coordinates, and stroke information, were used directly without pre-processing.

c) Feature Extraction

Two types of features were extracted:

- **Static features** from the offline image:
 - **GLCM** and **HOG (Histogram of Oriented Gradients)** were used to extract texture features.
 - **Geometric features**, including area, axis lengths, perimeter, and solidity, were calculated to describe global properties.
- **Dynamic features** from online data:
 - Features such as **velocity**, **acceleration**, **pen angle**, and **curvature radius** were calculated from pen movement data, enhancing the model's ability to detect skilled forgeries.

d) Classification

Each feature set was sent to a separate classifier:

- **SVM (Support Vector Machine)** was used to classify static image features.
- **DTW (Dynamic Time Warping)** was used for dynamic time-series data.

Each classifier produced a score indicating the likelihood of a signature being genuine.

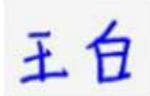
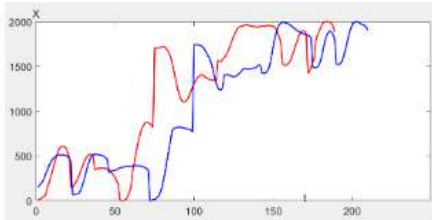
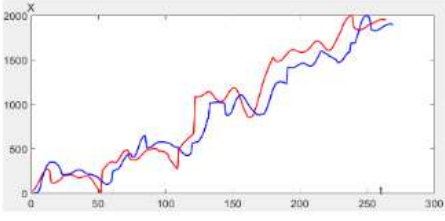
e) Score Fusion – SF-A

To combine the two scores from SVM and DTW, the study introduced a **Score Fusion based on Accuracy (SF-A)** method. It assigns a weight to each classifier's score based on its accuracy. The final decision is made by a weighted sum of both scores:

- If the final combined score is below zero, the signature is classified as genuine.
- If the score is above zero, it is considered forged.

This fusion technique improves robustness by utilizing the strengths of both classifiers and compensating for their individual weaknesses.

Table 1. Signature data.

		Genuine	Forgery
Writer1	Offline		
	Online		
Writer2	Offline		
	Online		

4. Findings

The experiments were conducted using different numbers of training samples (3, 5, 8, and 10 genuine signatures per person) to test the model's adaptability in low-data scenarios. The main performance metrics included:

- **False Acceptance Rate (FAR):** Incorrectly accepting forged signatures.
- **False Rejection Rate (FRR):** Incorrectly rejecting genuine signatures.
- **Average Error Rate (AER):** The mean of FAR and FRR.
- **Accuracy:** Percentage of correctly classified signatures.

Performance with Three Training Samples (Table 4)

This scenario simulates an environment where only a very small number of genuine signatures are available for training.

- **SVM using GLCM features** achieved 81.17% accuracy with a FAR of 16.83% and FRR of 20.83%.
- When combining GLCM and HOG features, accuracy improved to 85.17%, showing that combining local and directional features yields better results.
- Geometric features alone underperformed with only 67.08% accuracy, indicating that geometric data lacks sufficient variability for discrimination.
- **DTW**, used for dynamic features, alone achieved a respectable 89.17% accuracy.
- The **SF-A fusion method**, which combines the decision scores of SVM and DTW based on accuracy weighting, produced the highest accuracy of **93.08%**, with an AER of only 6.92%. This clearly demonstrates that combining static and dynamic features significantly improves performance, even when only three samples are used for training.

Table 4. The result of three training samples.

Method	Features	FAR	FRR	AER	Accuracy
SVM	GLCM	16.83%	20.83%	18.83%	81.17%
	HOG	24.83%	21.83%	23.33%	76.67%
	GLCM + HOG	12.17%	17.50%	14.83%	85.17%
	Geometric	37.83%	28.00%	32.92%	67.08%
	Texture + Geometric	13.00%	17.50%	15.25%	84.75%
DTW		11.17%	10.50%	10.84%	89.17%
SF-L		7.17%	7.67%	7.42%	92.58%
SF-A		6.67%	7.17%	6.92%	93.08%

Performance with Five Training Samples (Table 5)

With a slightly increased number of genuine training samples, overall accuracy improved across all classifiers.

- **SVM with GLCM + HOG** features showed a marked improvement with 90.50% accuracy.
- Combining texture and geometric features also performed well with 90.75% accuracy.
- **DTW alone** provided 90.42% accuracy, confirming its reliability in dynamic feature processing.
- **SF-A again outperformed all individual methods**, achieving **94.92% accuracy** and reducing AER to just 5.08%. Compared to SF-L (logistic regression-based fusion), SF-A consistently provided better results, due to its intelligent weighting based on classifier performance.

Table 5. The result of five training samples.

Method	Features	FAR	FRR	AER	Accuracy
SVM	GLCM	14.50%	13.17%	13.83%	86.17%
	HOG	11.83%	19.33%	15.58%	84.42%
	GLCM + HOG	6.67%	12.33%	9.50%	90.50%
	Geometric	29.17%	26.50%	27.83%	72.17%
	Texture + Geometric	6.33%	12.17%	9.25%	90.75%
DTW		9.83%	9.33%	9.58%	90.42%
SF-L		5.33%	5.50%	5.42%	94.58%
SF-A		5.33%	4.83%	5.08%	94.92%

Performance with Eight Training Samples (Table 6)

This scenario reflects a more realistic case where a moderate number of samples are available.

- SVM with HOG features alone provided **91.83%**, while GLCM + HOG achieved **94.33%**.
- DTW alone yielded **92.25%**, consistent with its earlier performance.
- **SF-L** (logistic regression fusion) achieved 97.25%, while ****SF-A** slightly surpassed it with an accuracy of **97.33%** and a very low AER of **2.67%**.

Table 6. The result of eight training samples.

Method	Features	FAR	FRR	AER	Accuracy
SVM	GLCM	10.83%	10.67%	10.75%	89.25%
	HOG	8.17%	8.17%	8.17%	91.83%
	GLCM + HOG	5.50%	5.83%	5.67%	94.33%
	Geometric	20.00%	27.33%	23.67%	76.33%
	Texture + Geometric	5.67%	5.83%	5.75%	94.25%
DTW		8.00%	7.50%	7.75%	92.25%
SF-L		3.67%	1.83%	2.75%	97.25%
SF-A		2.17%	3.17%	2.67%	97.33%

5. Conclusion

This research proposes a practical and effective method for handwritten signature verification that leverages the benefits of both offline and online data. By introducing a **score fusion technique (SF-A)** based on classifier accuracy, the system intelligently merges decision scores from SVM and DTW to deliver reliable and precise verification results.

One of the standout features of this approach is its **suitability for real-world applications**, where collecting large volumes of training data may not be feasible. By using as few as 3–5 genuine signature samples, the system still performs with high accuracy, making it ideal for scenarios such as banking and mobile authentication.

Additionally, the use of smart pen technology to gather rich offline and online data in a single stroke represents a valuable innovation. It enables deeper insight into signature behavior, including pressure patterns and stroke dynamics, which are hard to replicate in forgeries.

In future work, the authors suggest expanding the dataset to include more languages and scripts, which would help validate the generalizability of the method. Publicly releasing the dataset and source code could also encourage further research and innovation in the field of biometric verification.

Overall, this work offers a **balanced, scalable, and accurate approach** to the growing challenge of handwritten signature verification, aligning well with current advancements in artificial intelligence, pattern recognition, and real-time biometric security.